FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Normal Labour: Diagnosis, Mechanism, Progress & Delivery

Lecture | Week 2 · Labour, delivery & the puerperium | Obstetrics | Dr Nanzira

Labour is regular painful contractions that cause progressive cervical effacement and dilatation. Diagnose it correctly, track it on the partograph or Labour Care Guide, understand the eight cardinal movements, conduct a clean delivery and always give active management of the third stage (oxytocin 10 IU IM within 1 minute) and immediate newborn care.

Must know before the ward round

1. Labour = regular painful contractions + progressive cervical change. Contractions without cervical change are false labour (or latent phase), not an emergency.
2. WHO 2018 - active first stage starts at 5 cm; the older modified WHO partograph (still widely used in Uganda) starts active phase at 4 cm with alert line at 1 cm/hour.
3. Admission: first 5 minutes = pulse, BP, temperature, FHR after a contraction, and ask about bleeding, liquor colour, fetal movements and urge to push.
4. Adequate contractions = 3-5 in 10 minutes lasting 40-60 s; >5 in 10 minutes is tachysystole. Normal FHR 110-160 bpm.
5. Always write descent in fifths next to station - a head 3/5 palpable with "station +2" is caput, not descent.
6. Slow progress in a multipara with good contractions is obstruction until proved otherwise - no oxytocin; catheterise, rehydrate, senior review.
7. Cardinal movements - Engagement, Descent, Flexion, Internal rotation, Extension, Restitution, External rotation, Expulsion (Every Decent Fetus Is Exceptionally Rotated Every Expulsion).
8. Well-flexed vertex presents the suboccipitobregmatic diameter (9.5 cm) - the smallest - which is why occipito-anterior labour is normal labour.
9. AMTSL - oxytocin 10 IU IM within 1 minute of birth (after excluding a second twin), controlled cord traction with counter-traction, then uterine tone checks/massage every 15 minutes for 2 hours.
10. If oxytocin is unavailable - misoprostol 600 micrograms orally (Uganda Clinical Guidelines); never give ergometrine to a woman with hypertension or pre-eclampsia.
11. Delay cord clamping 1-3 minutes in a vigorous baby; dry, skin-to-skin, breastfeed within 1 hour, vitamin K 1 mg IM, tetracycline eye ointment, chlorhexidine cord care.
12. Second stage - WHO allows up to about 3 hours in first labours and 2 hours in subsequent labours if mother and baby are well; FHR every 5 minutes. Retained placenta = not delivered by 30 minutes.

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1. Definitions and Key Concepts

Labour is the process by which the fetus, placenta and membranes are expelled from the uterus after 28 weeks (Uganda's threshold of viability; WHO uses 22 weeks for statistics). **Normal labour** is spontaneous in onset, at term (37 to 42 weeks), with a **single fetus in vertex (occiput) presentation**, progressing without intervention to a spontaneous vaginal birth, with a healthy mother and baby afterwards.

> **CLINICAL PEARL**
> The one thing never to forget: **every birth gets a uterotonic within one minute** (after you have palpated the abdomen to exclude another baby). Postpartum haemorrhage is the leading cause of maternal death in Uganda, and active management of the third stage (AMTSL) roughly halves it.

Term	Definition (WHO 2018 unless stated)
Labour	Regular, painful uterine contractions leading to progressive effacement and dilatation of the cervix
False labour (Braxton Hicks)	Irregular contractions, usually painless or felt in the lower abdomen, relieved by rest/walking, **no cervical change**
Latent first stage	Painful contractions with variable cervical change, including some effacement and slow dilatation **up to 5 cm**
Active first stage	Regular painful contractions, substantial effacement, faster dilatation **from 5 cm to full dilatation (10 cm)**
Second stage	From full dilatation to birth of the baby; the woman has an involuntary urge to bear down. Divided into *passive* (before bearing down) and *active* (expulsive) phases
Third stage	From birth of the baby to delivery of the placenta and membranes
"Fourth stage"	The first 1 to 2 hours after placental delivery, when PPH risk is highest (not an official stage, but taught and examined)
Term	37+0 to 41+6 weeks
Engagement	Widest transverse diameter of the presenting part (biparietal, 9.5 cm) has passed through the pelvic inlet. Clinically: **2/5 or less palpable abdominally**
Station	Level of the lowest bony part of the head relative to the ischial spines (0 = at the spines; -3 to +3 cm)
Effacement	Shortening/thinning of the cervix as it is taken up into the lower segment
Crowning	The head no longer recedes between contractions (the biparietal diameter has passed the vulval ring)
Restitution	The return of the head to its natural alignment with the shoulders after delivery of the head (a 45-degree turn back towards the side where the occiput started)

Duration of normal labour

Stage	First labour (nullipara)	Subsequent labours (multipara)
Latent first stage	Variable; no standard length (can be 20 h or more)	Variable (can be 14 h or more)
Active first stage (from 5 cm)	Median about 4 h; usually not longer than 12 h	Median about 3 h; usually not longer than 10 h
Second stage	Usually up to 3 h	Usually up to 2 h
Third stage	5-15 min with AMTSL; over 30 min = prolonged (retained placenta)	Same

The old rule "cervix must dilate at 1 cm per hour" is the basis of the partograph alert line. WHO 2018 says a slower rate **alone** is not an indication to intervene if mother and baby are well, because many normal women dilate more slowly, especially before 5 cm.

> **UGANDA CONTEXT**
> The **modified WHO partograph** (used in many Ugandan units and in the Uganda Clinical Guidelines) defines the active phase from **4 cm**. The **WHO Labour Care Guide (LCG)**, adopted by the Uganda Ministry of Health with the Essential Maternal and Newborn Clinical Care guidelines (2022), starts at **5 cm**. Know both; use whichever form your ward uses, and say which one you are using when you present.

2. Burden and Uganda Context

Why it matters in Uganda: most maternal deaths (haemorrhage, obstructed labour, sepsis, ruptured uterus) and a large share of newborn deaths (birth asphyxia) happen **during labour and the first 24 hours after birth**. Almost every one of these starts as a normal labour that was not watched properly, or a problem that was not recognised on time. A junior clerk who can *diagnose labour, follow its progress on a partograph/Labour Care Guide, deliver the baby safely, give oxytocin in the first minute, and dry and assess the newborn* is directly preventing deaths.

- Uganda records about **1.5 million births a year**. Maternal mortality has fallen but remains high (UDHS 2022 estimated roughly 189 per 100,000 live births), and neonatal mortality is about 22 per 1,000 live births.
- Most births now happen in health facilities, but many are in **HC III and HC IV** units staffed by one or two midwives at night, with intermittent electricity, few Dopplers, and limited blood.
- The **"three delays"** model explains many deaths: delay in deciding to seek care, delay in reaching care (transport, money, night), and delay in receiving adequate care at the facility (no staff, no partograph filled, no oxytocin, no theatre).
- **Obstructed labour** remains a major cause of ruptured uterus, obstetric fistula (VVF/RVF) and fresh stillbirth in Uganda, and it is **preventable by careful monitoring and timely referral**.
- **Level of care:** HC II has no delivery service in principle; **HC III** provides basic emergency obstetric and newborn care (BEmONC: uterotonics, $MgSO_4$, antibiotics, manual removal of placenta, MVA, assisted vaginal delivery, newborn resuscitation). **HC IV** and hospitals provide **comprehensive EmONC** (add caesarean section and blood transfusion).

Setting: who does what where

Level	What can be done
HC III	Normal delivery, partograph/LCG, AMTSL, newborn resuscitation with bag and mask, loading dose $MgSO_4$, first dose antibiotics, manual removal of placenta, **early referral**
HC IV / general hospital	All of the above plus caesarean section, blood transfusion, oxytocin augmentation with close monitoring, vacuum delivery, SCBU (some)
Regional/national referral (e.g. Mengo, Mulago, Kawempe)	Specialist obstetric and neonatal care, ICU, high-risk cases

> **UGANDA CONTEXT**
> A woman labouring at an HC III with a slow partograph needs **early referral** (when the alert line is crossed), because transport to the HC IV/hospital can take hours. Waiting for the action line at an HC III is too late.

3. Principles and Framework

The 3 Ps (plus 2)

Progress in labour depends on the **Powers** (contractions), the **Passenger** (fetus: size, lie, presentation, position, attitude) and the **Passage** (bony pelvis and soft tissues). Add the **Psyche** (fear increases catecholamines, which slows labour; a companion helps) and the **Provider** (good support vs unnecessary interventions).

Powers: uterine contractions

- The uterus is smooth muscle. Near term, **oxytocin receptors, gap junctions (connexin-43) and prostaglandin receptors** increase sharply, so the myometrium changes from quiet to coordinated.
- **Progesterone withdrawal** (functional, in humans) and a rise in **oestrogen**, **CRH from the placenta**, fetal cortisol and **prostaglandins (PGE2, PGF2alpha)** from membranes and decidua all push the uterus into labour.
- **Ferguson reflex**: stretching of the cervix and upper vagina by the presenting part triggers oxytocin release from the posterior pituitary, which strengthens contractions. This is a positive feedback loop, and it explains why labour accelerates once the head is well applied to the cervix, and why a poorly applied head (malpresentation, high head, big CPD) often gives poor contractions.
- **Fundal dominance**: contractions start near the cornua and spread downwards; they are strongest and longest in the fundus. The upper segment **contracts and retracts** (the fibres become permanently shorter after each contraction), and the passive lower segment stretches and thins. This pulls the cervix up (effacement) and open (dilatation).
- **Pathological retraction ring (Bandl's ring)**: in obstructed labour the upper segment keeps retracting against an immovable fetus, the lower segment becomes dangerously thin, and a ridge becomes visible across the abdomen. This is a warning of **imminent uterine rupture**.
- **Adequate contractions** in the active phase: 3-5 in 10 minutes, each lasting 40-60 seconds. More than 5 in 10 minutes (**tachysystole**) reduces placental blood flow (the placental bed is only perfused between contractions) and causes fetal distress.

The passenger: fetal skull

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    Anterior fontanelle (bregma) - diamond, 4 sutures meet
          \
 frontal  \   parietal        parietal
 bone ----[ ]-------------------------  occiput
           ^ sagittal suture           ^
                                        Posterior fontanelle (lambda)
                                        - Y/triangle, 3 sutures meet

Diameters (anteroposterior) and the attitude that presents them:
Suboccipitobregmatic   9.5 cm   well-flexed vertex  -> normal labour
Suboccipitofrontal    10.0 cm   partly flexed vertex
Occipitofrontal       11.5 cm   deflexed (military) -> OP positions
Mentovertical         13.0 cm   brow  -> will NOT deliver vaginally at term
Submentobregmatic      9.5 cm   face (mento-anterior can deliver)
Transverse: biparietal 9.5 cm, bitemporal 8.0 cm
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The **denominator** of a vertex is the **occiput**. Position describes where the occiput points: left occipito-anterior (LOA), direct OA, right OA, left/right occipito-transverse (LOT/ROT), and occipito-posterior (LOP/ROP/direct OP).

Moulding: the parietal bones override each other and the occipital/frontal bones slide under them, reducing the diameters a little. **Graded 0 (bones separate), + (touching), ++ (overlapping but reducible), +++ (overlapping, not reducible)**. **Caput succedaneum** is scalp oedema over the presenting part. Severe moulding plus big caput

plus a head still high = **cephalopelvic disproportion (CPD)**.

The passage: the pelvis

Plane	Widest diameter	Shape
Inlet (brim)	**Transverse** (about 13 cm); AP (obstetric conjugate) about 11 cm	Oval side-to-side
Mid-cavity	Round (about 12 cm)	Round
Outlet	**Anteroposterior** (about 12.5-13 cm with coccyx moved back); transverse (intertuberous) about 11 cm	Oval front-to-back

Because the inlet is widest **side to side** and the outlet is widest **front to back**, the head must **enter transverse and rotate to anterior** to fit. That is the whole logic of the mechanism of labour. The **gynaecoid** pelvis is ideal; **android** pelvis (narrow forepelvis, prominent spines) favours deep transverse arrest and OP positions.

The mechanism of labour (cardinal movements) for LOA

1. **Engagement**: head enters the brim in the **transverse or oblique** diameter, usually by 36 weeks in primigravidae (often only in labour in multiparas). Clinically, 2/5 or less palpable above the brim.
2. **Descent**: continues throughout labour, driven by contractions, and in the second stage by maternal bearing down.
3. **Flexion**: as the head meets resistance from the cervix, pelvic walls and pelvic floor, the chin tucks onto the chest, changing the presenting diameter from occipitofrontal (11.5 cm) to **suboccipitobregmatic (9.5 cm)**. The occiput becomes the leading part.
4. **Internal rotation**: the leading occiput hits the gutter-shaped **pelvic floor (levator ani)**, which directs it forwards and medially; the occiput rotates **one-eighth of a circle (45 degrees)** from LOA to OA, so the sagittal suture lies in the AP diameter of the outlet. The shoulders are still oblique, so the neck twists slightly. (Rule: "whatever part reaches the pelvic floor first rotates forwards.")
5. **Extension (crowning and birth of the head)**: the nape of the neck pivots under the pubic symphysis (suboccipital region) and the head extends: occiput, then bregma, forehead, face and chin sweep over the perineum. **Crowning** = the head no longer recedes between contractions (the biparietal diameter has passed the vulval ring).
6. **Restitution**: once the head is free, the neck untwists and the occiput turns back 45 degrees to the side it came from (LOA baby: occiput turns to the mother's left).
7. **External rotation**: the anterior shoulder reaches the pelvic floor and rotates forwards to lie under the symphysis; the head turns a further 45 degrees, so the face looks at the mother's thigh. The bisacromial diameter (12 cm) is now in the AP of the outlet.
8. **Expulsion**: the anterior shoulder is born under the symphysis by gentle downward (posterior) traction, then the posterior shoulder by upward (anterior) traction, then the trunk by lateral flexion.

EXAM FAVOURITE

Viva favourite: "Why does the head rotate?" Answer: **the shape of the pelvic floor** (a forward-sloping gutter formed by the levator ani) and the **shape of the pelvis** (outlet widest AP) make the leading, well-flexed occiput rotate anteriorly. A lax pelvic floor (grand multipara) or a deflexed head (OP) is why rotation fails.

Third stage physiology

After the baby is born, the uterus contracts down sharply. The placental site shrinks but the placenta cannot, so it **shears off** the decidua spongiosa; a retroplacental clot forms. Signs of separation: (1) **the uterus becomes firm, globular and rises** in the abdomen, (2) **a gush of blood**, (3) **lengthening of the cord** at the introitus. Haemostasis depends on the **"living ligatures"**: the criss-cross myometrial fibres kink the spiral arteries shut when the uterus contracts. This is why **atony = bleeding**, and why an oxytocic is the core of PPH prevention.

The framework for every review

Ask the four questions every time you review a labouring woman: *Is the mother well? Is the baby well? Is labour progressing? Is the partograph/LCG up to date?*

4. Application in Practice

This is the **admission-in-labour workup** as it is done at the admission desk of a Ugandan labour ward (Mengo, Mulago/Kawempe, or a busy HC IV), followed by how you **reassess progress** at every review, then the conduct of each stage. The admission takes 15–20 minutes for a well woman, and the first 5 minutes decide whether she is a routine admission or an emergency.

Step 0: the first 5 minutes

Before the detailed history, do a rapid **"is anyone in danger?"** check, because women often arrive late, after hours of labour at home or at a lower facility.

1. **Look at her**: is she pushing, bleeding, fitting, pale, breathless, exhausted or confused? Is there a **cord or a limb at the introitus**?
2. **Vital signs at once**: pulse, BP, temperature, respiratory rate. Calculate the **shock index** (pulse ÷ systolic BP): **≥0.9 is abnormal** and needs escalation.
3. **Fetal heart**: listen for a full minute immediately after a contraction with a Pinard or Doppler. Normal is **110–160 bpm**.
4. **Quick abdominal look**: scar? Bandl's ring? Full bladder? Obvious transverse lie?
5. **Ask four questions**: "Are you bleeding? Has water broken, and what colour? Is the baby moving? Do you feel like pushing?"

The actions for abnormal findings are in section 6 (first-5-minute emergencies).

Step 1: focused history (each answer changes the plan)

Question	Why it matters
Age, gravida, para, **gestational age** (LNMP, early scan, SFH)	Term vs preterm: <37 weeks → preterm labour pathway (steroids if 28–34 weeks, tocolysis, neonatal team). Grand multiparity (P5+) = PPH and uterine rupture risk; oxytocin augmentation only after senior review
When did pains start? How often? How long? Are they getting stronger?	Distinguish true vs false labour; estimate how long she has really been labouring (prolonged labour at home is common in Uganda; hours at home count)
Has there been a show or water breaking? When? Colour of the liquor?	Rupture of membranes **>18 h** = infection risk (GBS prophylaxis); **meconium** = possible fetal compromise, closer monitoring; **blood** = abruption or placenta praevia
Is there vaginal bleeding (more than a show)?	Antepartum haemorrhage: **never do a digital VE until praevia is excluded**

Question	Why it matters
Is the baby moving normally?	Reduced movements may mean fetal compromise: check FHR at once; absent FHR = IUFD pathway
Headache, blurred vision, epigastric pain, fits?	Pre-eclampsia/eclampsia
Fever, foul discharge, burning urine?	Chorioamnionitis, UTI, malaria (needs antibiotics or antimalarials)
Previous deliveries: mode, **previous caesarean or myomectomy/uterine surgery**, birth weights, shoulder dystocia, PPH, retained placenta, perineal tears (3rd/4th degree), stillbirths, duration	Previous CS = trial of labour after CS (TOLAC) only at a facility with theatre; past big baby/shoulder dystocia/PPH predicts recurrence
ANC card: number of visits, **HIV status** (ART, last viral load), syphilis, Hb, blood group/Rh, sickle status, IPTp, Td doses, scan reports (placenta site, number of babies)	HIV-positive: continue ART, avoid unnecessary ARM/episiotomy, give infant prophylaxis. Anaemia lowers tolerance to blood loss
Chronic illness and drugs (hypertension, diabetes, cardiac disease, sickle cell, asthma)	Affects choice of uterotonic (hypertension: no ergometrine), fluids (cardiac: careful), analgesia, monitoring
Referral note if referred in	Time of referral, drugs given ($MgSO_4$, antibiotics, oxytocin), and the partograph from the referring unit
Birth companion, birth plan, family planning intentions	Respectful maternity care; PPFP (e.g. IUD within 48 h, implant) is offered in labour ward

Step 2: focused examination (Hutchison order, obstetric abdomen)

1. **Introduction and consent**: introduce yourself, confirm her identity, explain, gain consent, **chaperone** for the VE. Respectful maternity care is a WHO standard, and a relaxed woman labours better (less adrenaline).
2. **Hand hygiene.**
3. **Position and exposure**: semi-recumbent with a left tilt, between contractions; expose the abdomen, keep her covered otherwise; ask her to empty her bladder.
4. **General inspection**: general condition, distress, exhaustion, **pallor**, jaundice, **hydration** (dry tongue, sunken eyes, ketones), oedema.
5. **Vital signs**: **pulse, BP**, temperature, respiratory rate. Tachycardia above 100 with fever suggests infection or dehydration; above 120 is an alert.
6. **Hands, face and neck**: palmar and conjunctival pallor, facial oedema.
7. **Abdominal examination** (between contractions):

Step	Normal finding to document	Abnormal finding that changes the plan
Inspection	Longitudinal ovoid, no scars, fetal movements	CS scar, **Bandl's ring**, distended bladder, "hour-glass" shape
Fundal height (SFH)	Consistent with dates	Large for dates (twins, polyhydramnios, macrosomia); small (FGR)
Lie	Longitudinal	Transverse or oblique
Presentation	Cephalic	Breech
Position (where is the back?)	Back on one side (LOA/ROA)	Limbs all over the front (OP)
Engagement / descent in fifths (5/5 = floating; 0/5 = fully in the pelvis)	3/5 or less and falling	5/5 or 4/5 in active labour with strong contractions (CPD)

Step	Normal finding to document	Abnormal finding that changes the plan
Contractions (palpate for 10 min: number, and duration from hardening to relaxing)	3–5 in 10 min, each 40–60 s	≤2 (inadequate) or >5 (tachysystole); continuous pain (abruption)
Auscultation of the fetal heart (Pinard or Doppler, full minute, **immediately after a contraction**)	110–160 bpm, regular	<110, >160, slowing after contractions
Tenderness	None between contractions	Tender lower segment (obstruction), tender tense uterus (abruption)

Descent in fifths **is more useful than station** because caput can make station look low when the head is actually high.

8. **Vaginal examination** (aseptic, sterile gloves, after consent, with a chaperone; **not if there is APH** until praevia is excluded):
    1. Vulva: discharge, liquor colour, bleeding, warts/ulcers, FGM scarring, oedema.
    2. **Cervix**: position, consistency, **effacement** (length in cm or %), **dilatation** (cm), application to the presenting part, oedema.
    3. **Membranes**: intact or ruptured; **liquor** colour (clear/meconium/blood-stained).
    4. **Presenting part**: vertex, face, brow, breech; **exclude cord presentation or prolapse**.
    5. **Position** by sutures and fontanelles (feel the posterior fontanelle: small Y-shape, anterior = OA).
    6. **Station** relative to the ischial spines, **caput** and **moulding**.
    7. **Pelvis** (clinical pelvimetry): sacral promontory reachable? spines prominent? subpubic angle (admits 2 fingers normally)? intertuberous diameter (4 knuckles)?
9. **Complete**: cover her, thank her, **hand hygiene**, document positives and negatives, start or update the partograph/LCG.

> CLINICAL PEARL
> Always write **descent in fifths (abdomen) next to station (VE)**. "Cervix 6 cm, head 3/5, station 0, caput ++, moulding ++" tells a senior that the "low" head on VE is mostly caput: think CPD.

Step 3: bedside tests

Test	How / when	What you do with it
Partograph / LCG	Start in the **active phase** (4 cm on the modified WHO partograph; 5 cm on the LCG)	The core bedside "test" of labour progress; see the partograph topic
Intermittent auscultation (Pinard / hand-held Doppler)	Every 30 min in the active first stage (WHO: 15–30 min); every 5 min in the second stage	Abnormal FHR → intrauterine resuscitation, VE, senior review
Urine dipstick	On admission and each void	Protein ≥2+ with BP ≥140/90 → pre-eclampsia workup; ketones → oral/IV fluids; nitrites/leucocytes + fever → UTI
HemoCue / Hb	On admission if no recent Hb	Hb <11 → anaemia; **<7 severe** → group and crossmatch, deliver in a unit with blood
HIV rapid test	If status unknown, or last negative test >3 months ago	Positive → start TLD now, infant prophylaxis, avoid unnecessary ARM/episiotomy
Syphilis (dual HIV/syphilis RDT)	If not done in ANC	Reactive → benzathine penicillin; examine and treat the baby

Test	How / when	What you do with it
Malaria RDT	If febrile	Positive → treat malaria; fever may still be chorioamnionitis
Random blood sugar (glucometer)	Diabetes, prolonged fasting, vomiting	Hypoglycaemia → dextrose; hyperglycaemia → sliding scale per protocol
Bedside clotting test	Suspected abruption, IUFD, PPH	Clot not formed by 7 min (or breaks down easily) → coagulopathy; fresh blood/FFP
Bladder check by palpation / catheter	Suprapubic fullness, slow descent	Empty the bladder, then reassess descent

Step 4: laboratory and imaging

Normal labour is a **clinical** diagnosis; the lab is for risk and safety, not to confirm labour.

Test	When	Expected / abnormal finding	How it changes management
Hb / FBC	All admissions without a recent Hb; anyone with bleeding, pallor, fever	Hb ≥11 g/dL; WBC up to ~25 × 10^9/L is normal in labour. Hb <7 = severe anaemia; platelets <100 = think HELLP/DIC	Severe anaemia: crossmatch, deliver where blood is available, active PPH prevention. Available at HC IV (HemoCue) and all hospitals
Blood group & Rh, crossmatch	Not on ANC card; any high-risk woman (previous CS, APH, grand multipara, anaemia)	Rh negative; antibodies	Rh negative → anti-D within 72 h if baby Rh positive (cord blood group). Crossmatch at HC IV/hospital; HC III must refer
Cord blood group (baby of an Rh-negative mother)	At birth	Baby Rh positive	Decides whether the mother needs anti-D within 72 hours
HIV and syphilis RDT	Unknown status or retest due	Reactive	ART, infant prophylaxis; penicillin for mother and baby. Available at all levels (HC III+)
Repeat HIV test in labour	Last negative test >3 months ago, or in the third trimester in high-risk women	Recent seroconversion	Acute infection carries the highest transmission risk; start ART and enhanced infant prophylaxis
Urinalysis (dipstick) ± microscopy/culture	Admission; fever or dysuria	Proteinuria, ketones, nitrites	Pre-eclampsia pathway; hydration; antibiotics. Dipstick everywhere; culture only at regional/national referral
Malaria RDT / blood smear	Fever	Positive	Antimalarial (IV artesunate if severe). RDT available at all levels
Blood glucose	Diabetes, fasting, IUFD	Hypo- or hyperglycaemia	Dextrose or insulin; neonatal glucose monitoring
LFTs, creatinine, platelets	BP ≥140/90 with symptoms or proteinuria	Raised ALT/AST, creatinine >90 µmol/L, low platelets	Severe pre-eclampsia/HELLP → $MgSO_4$ and delivery plan. Regional/national referral hospitals
Bedside ultrasound	Uncertain presentation, suspected twins, no FHR heard, APH (placenta site)	Cephalic singleton with FHR; or breech/twins/IUFD/praevia	Changes mode of delivery. Available at most HC IVs and hospitals (depends on a trained person)

Test	When	Expected / abnormal finding	How it changes management
CTG	High-risk labour only (not routine for low-risk women, WHO 2018)	Normal vs suspicious/pathological trace	Available at Mengo, Mulago/Kawempe and some regional referrals; not at HC III/IV

Step 5: first-stage care

1. **Decide**: false labour, latent phase, or active labour? Normal or high risk?
2. **Start the partograph/LCG** once she is in the active phase (4 cm on the modified WHO partograph; 5 cm on the LCG). Record latent-phase findings in the notes.
3. **Supportive care**:
   - **Birth companion** of her choice.
   - **Oral fluids and light food** (do not starve low-risk women; dehydration and ketosis impair contractions). If vomiting or dehydrated: IV Ringer's lactate or normal saline.
   - **Mobility and upright positions**; avoid lying flat on the back (aortocaval compression reduces placental blood flow).
   - **Empty the bladder** every 2 hours (a full bladder impedes descent and causes atony later).
   - **Pain relief** on request: non-drug methods (massage, warm compress, breathing, positions); pethidine 50-100 mg IM (with an antiemetic) in the early active phase if needed, **avoid within about 2-3 hours of expected birth** because it depresses the newborn's breathing. Epidural is available only in a few Ugandan hospitals.
4. **Monitoring** (modified WHO partograph schedule):

Parameter	Frequency in active first stage	Frequency in second stage
Fetal heart rate (after a contraction, 1 full minute)	Every 30 min (WHO 2018: every 15-30 min)	Every 5 min (or after every contraction)
Contractions (number in 10 min, duration)	Every 30 min	Continuously
Maternal pulse	Every 30 min	Every 5-15 min
BP	Every 4 h (more often if abnormal)	Every 15-30 min
Temperature	Every 2 h (LCG: every 4 h)	
Urine (volume, protein, ketones)	Every time passed	
Vaginal examination (cervix, descent, liquor, moulding)	Every 4 h, or sooner if a clinical indication	When the urge to push begins or to confirm full dilatation

5. **Avoid routine interventions** in a normal labour: no routine enema, pubic shaving, routine amniotomy, routine IV fluids, routine oxytocin, or routine CTG (WHO 2018).
6. **Infection prevention**: hand hygiene, sterile gloves for each VE, minimise VEs, especially after rupture of membranes.

Step 6: second stage: conducting the delivery

Diagnose full dilatation by VE (or visible head at the introitus). Signs: urge to push, anal pouting, perineal bulging, the head visible.

1. **Prepare**: delivery pack open on a clean trolley, **oxytocin 10 IU drawn up** (check for a second twin before giving), newborn corner ready (warm room, 2 dry towels, cap, bag and mask size 1 and 0 checked, suction bulb, clock), gloves, apron, goggles.

2. **Position**: the position she prefers (semi-sitting, left lateral, squatting, kneeling). WHO encourages upright positions.
3. **Pushing**: let her push **with her own urge** rather than coached prolonged Valsalva. In a primigravida with an epidural, allow a passive phase of 1-2 hours before pushing if the FHR is fine.
4. **Monitor FHR every 5 minutes** or after each contraction; pulse every 15 minutes.
5. **Perineal protection**: warm compresses, perineal massage in second stage, and "hands-on" control of the head at crowning. Ask her to **pant or give short gentle pushes** as the head crowns, so it is born slowly between contractions (sudden expulsion tears the perineum).
6. **Episiotomy only if indicated** (fetal distress needing rapid delivery, instrumental delivery, rigid perineum causing delay, shoulder dystocia manoeuvres). Not routine (WHO 2018). Mediolateral, at about 60 degrees from the midline at crowning, after infiltrating 1% lidocaine.
7. **Delivery of the head**: keep the head flexed with the left hand on the occiput; the right hand supports the perineum with a pad. Let the head extend slowly.
8. **Wipe the face** (do not routinely suction). **Feel for cord around the neck**: if loose, slip it over the head; if tight, clamp twice and cut between (rarely needed; do this only if it prevents delivery).
9. **Wait for restitution and external rotation** (the next contraction).
10. **Deliver the anterior shoulder** by gentle downward traction on the head (hands on the sides of the head, over the parietal bones, never pulling on the neck), then the **posterior shoulder** by lifting upwards, then the body by lateral flexion towards the mother's abdomen.
11. **Note the time of birth** and sex, place the baby on the mother's abdomen.

Step 7: immediate newborn care (the first minute, then the first hour)

1. **Dry thoroughly** (head, body, limbs) with a warm dry towel within the first 30 seconds while assessing breathing; remove the wet towel.
2. **Crying or breathing well?**
   - **Yes**: keep **skin-to-skin** on the mother's chest, cover with a dry cloth and cap. **Delay cord clamping for 1-3 minutes** (not earlier than 1 minute).
   - **No**: clear the airway **only if** secretions are blocking it, stimulate by rubbing the back, and if still not breathing **clamp and cut the cord and start bag-and-mask ventilation within the Golden Minute** (see the neonatal resuscitation topic).
3. **Give the mother oxytocin 10 IU IM** within 1 minute of birth (after palpating the abdomen to exclude a second baby).
4. **Cord**: clamp/tie at about 2 cm and 5 cm from the umbilicus and cut between with sterile scissors. Apply **chlorhexidine 7.1% digluconate gel** to the stump (Uganda policy since 2016), keep it clean and dry; nothing else on the cord.
5. **Breastfeeding within the first hour**, uninterrupted skin-to-skin for at least an hour.
6. **After the first hour**: weigh, full newborn examination, **vitamin K (phytomenadione) 1 mg IM** into the anterolateral thigh (**0.5 mg for preterm/low birth weight babies** per local protocol), **tetracycline 1% eye ointment** to both eyes once, identification band.
7. **Immunisation**: BCG and OPV-0 at birth (and hepatitis B birth dose where it has been introduced).
8. **Do not bathe for at least 24 hours** (hypothermia risk). Keep mother and baby together (rooming-in).

Newborn drug	Dose	Route	When	Notes
Vitamin K (phytomenadione)	1 mg (term); 0.5 mg preterm/LBW	IM, anterolateral thigh	Once, in the first hours (WHO: within 6 h)	Prevents vitamin K deficiency bleeding

Newborn drug	Dose	Route	When	Notes
Tetracycline eye ointment 1%	Thin ribbon each eye	Topical	Once, within 1 h	Prevents ophthalmia neonatorum (gonococcal/chlamydial)
Chlorhexidine 7.1% digluconate gel (4% chlorhexidine)	Apply to the stump and base	Topical	From birth, per local protocol	Reduces omphalitis and neonatal sepsis
BCG	0.05 mL	Intradermal, right upper arm	At birth	
OPV-0	2 drops	Oral	At birth (within 2 weeks)	

Step 8: third stage: active management (AMTSL)

The three components, in order:

1. **Uterotonic within 1 minute of birth** (after excluding another fetus).
2. **Controlled cord traction (CCT)** when the uterus contracts, **with counter-traction** on the uterus (Brandt-Andrews manoeuvre). CCT is recommended where skilled attendants are present; if the provider is not skilled in CCT, wait for signs of separation and let the mother push the placenta out.
3. **Uterine tone assessment** (palpation) after the placenta, and fundal massage if soft; check every 15 minutes for 2 hours. Sustained continuous massage is no longer recommended when oxytocin has been given, but tone must be checked.

How to do CCT

1. Clamp the cord close to the perineum and hold it with one hand.
2. Place the other hand just above the pubic bone and **push the uterus upwards (towards the umbilicus)** during a contraction. This counter-traction prevents **uterine inversion**.
3. Apply **steady, gentle downward traction** on the cord during the contraction.
4. If the placenta does not descend within 30-40 seconds, stop, wait for the next contraction and repeat.
5. As the placenta appears, hold it with both hands and turn it gently to twist the membranes into a rope; tease them out slowly (sponge forceps if needed).
6. **Examine the placenta and membranes** for completeness (missing cotyledons, succenturiate lobe vessels running to the edge of the membranes), count the cord vessels (2 arteries + 1 vein).

Drug (PPH prevention)	Dose	Route	Frequency	Notes
Oxytocin (first choice)	**10 IU**	**IM** (or slow IV)	Once, within 1 min of birth	Store at 2-8 °C (cold chain). Acts in 2-3 min IM. Not a bolus IV push in shocked or cardiac patients
Misoprostol (if no oxytocin)	**600 micrograms**	**Oral**	Once	Uganda Clinical Guidelines dose; WHO allows 400 or 600 micrograms. Side effects: shivering, fever
Heat-stable carbetocin (where supplied)	100 micrograms	IM or IV	Once	WHO alternative; does not need cold chain; **not** for induction/augmentation
Ergometrine (or oxytocin-ergometrine)	0.2 mg (ergometrine)	IM	Once	**Contraindicated in hypertension, pre-eclampsia, cardiac disease**; causes sustained contraction; may retain placenta

DRUG DOSING

Always palpate the abdomen for a second twin before giving the oxytocin. Oxytocin given with an undiagnosed second twin in utero can make the uterus contract tonically around the second baby and asphyxiate it.

Step 9: the fourth stage (first 1-2 hours) and early postpartum care

1. **Inspect the perineum, vagina and cervix** for tears; repair (see the vaginal delivery skills topic).
2. **Estimate blood loss** (normal under 500 mL). Weigh soaked pads and drapes if needed, or use a calibrated drape.
3. **Monitor every 15 minutes for 2 hours**: uterine tone (and massage if soft), vaginal bleeding, pulse, BP. Then every 30 min for the next hour, then 4-hourly.
4. **Empty the bladder**: encourage voiding within 2-6 hours (a full bladder pushes up the uterus and causes atony).
5. **Keep mother and baby together**, observe the baby (breathing, colour, warmth, feeding).
6. **Counsel**: danger signs (heavy bleeding, fever, headache/fits, foul lochia, baby not feeding), exclusive breastfeeding, **postpartum family planning** (IUD within 48 h, implant, POPs; LAM), hygiene.
7. **Discharge** after at least **24 hours** for facility births (WHO), with postnatal visits at day 3, day 7-14 and 6 weeks.

5. Clinical Acumen

Red flags in labour (act and call for help)

RED FLAG

- Vaginal bleeding more than a show (abruption, praevia, ruptured uterus)
- FHR below 110 or above 160, or decelerations after contractions (late decelerations)
- Thick meconium-stained liquor
- Cord prolapse or cord presentation
- Maternal fever of 38 °C or more, maternal pulse above 120, BP 140/90 or more (or diastolic 110 or more), convulsions
- Partograph crossing the **action line**, or an LCG cervix alert (time spent at one dilatation reaches the LCG threshold: 6 h at 5 cm, 5 h at 6 cm, 3 h at 7 cm, 2.5 h at 8 cm, 2 h at 9 cm)
- Signs of obstruction: head not descending, severe moulding (+++), large caput, oedematous cervix/vulva, haematuria, **Bandl's ring**, maternal exhaustion and dehydration
- Signs of rupture: sudden severe pain, contractions stop, fetal parts easily felt, shock, FHR lost
- Malpresentation discovered in labour (breech, face, brow, transverse lie, compound)

RED FLAG

A multigravida arriving from home after a long labour, dehydrated, with a tender lower segment and a high head is **obstructed labour until proved otherwise**. Do not start oxytocin; rehydrate, catheterise, give antibiotics and prepare for caesarean section.

Thinking like a clinician

- **"Is she in labour?"** Only two things make the diagnosis: *regular painful contractions* and *cervical change on two examinations (or a cervix that is already 4-5 cm and effaced)*. A show and ruptured membranes support

it but do not define it.

- **"Is it normal?"** Ask the four questions every time you review a labouring woman: *Is the mother well? Is the baby well? Is labour progressing? Is the partograph/LCG up to date?*
- **If progress is slow, think the 3 Ps in order**: *Powers* (count contractions; inadequate? dehydrated? full bladder?), *Passenger* (malposition like OP, deflexion, big baby, malpresentation), *Passage* (CPD, contracted pelvis). Poor contractions in a multipara are rare; in a multipara slow progress is **obstruction** until proved otherwise, and oxytocin can rupture her uterus.
- **If X then think Y**:
  - Contractions strong, head not descending, moulding +++ in a primigravida → CPD/obstruction → caesarean, not oxytocin.
  - Maternal tachycardia + fever + tender uterus + foul liquor → chorioamnionitis → antibiotics, expedite delivery.
  - Suprapubic swelling that dulls on percussion → full bladder → catheterise, then reassess (a full bladder can stop descent).
  - Sudden fetal bradycardia after ARM → cord prolapse → VE immediately.
  - Lower abdominal ridge + tender lower segment + haematuria → impending rupture → stop oxytocin, emergency CS.
- **Common student mistakes**: doing VEs too often (infection) or not at all; plotting the first VE in the latent phase on the alert line; forgetting the FHR after a contraction; forgetting to palpate for a second twin before giving oxytocin; pulling on the cord before the uterus has contracted (uterine inversion); leaving the mother unobserved in the first hour after birth.

> CLINICAL PEARL
> The most common student error is calling a big caput "head at +2". Always **correlate station on VE with descent in fifths on abdominal palpation**. If the abdomen says 3/5 but the VE says +2, you are feeling caput, and the head is not engaged. That combination (high head + big caput + severe moulding) is obstruction until proved otherwise.

Differential diagnosis of "pains in late pregnancy"

Condition	Distinguishing features	Key investigation / action
True labour	Regular, increasing contractions; cervical dilatation/effacement progress; show	Serial VE 4 h apart; partograph/LCG
False labour / Braxton Hicks	Irregular, not increasing, relieved by rest/analgesia, cervix unchanged	Reassess after 2-4 h; discharge with advice if cervix unchanged and all well
Prolonged latent phase	Painful contractions, cervix less than 5 cm, slow change over many hours	Supportive care, analgesia, hydration; do not augment before 5 cm if mother/baby well
Preterm labour	Labour before 37 weeks	Gestational age dating; steroids (dexamethasone 6 mg IM 12-hourly × 4 doses if 28-34 weeks), tocolysis per protocol
Placental abruption	Continuous pain, tense tender "woody" uterus, dark bleeding, shock out of proportion, FHR abnormal or absent	Clinical; ultrasound may miss it; clotting (bedside clotting test)
Uterine rupture	Previous CS/grand multipara/obstructed labour; sudden pain, loss of contractions, shock, easily palpable fetal parts	Clinical; emergency laparotomy
Chorioamnionitis	Fever, maternal and fetal tachycardia, uterine tenderness, foul liquor	FBC, clinical; antibiotics

Condition	Distinguishing features	Key investigation / action
UTI/pyelonephritis, acute abdomen (appendicitis, torsion)	Pain not rhythmic with uterine hardening, dysuria, loin pain, localised tenderness	Urinalysis, ultrasound
Severe pre-eclampsia (epigastric pain)	BP 160/110, proteinuria, headache, RUQ pain	Urine protein, BP, platelets, LFTs

Putting it together

Model one-line summary: "A 24-year-old G2P1 at 39 weeks, HIV-negative, Hb 11.8, in **active first stage** (cervix 5 cm, head 2/5, LOA), mother and fetus well, with no risk factors."

Problem list for a typical admission: (1) active labour at term; (2) any risk factor you found (e.g. previous CS, anaemia, HIV, prolonged ROM, grand multiparity); (3) plan for AMTSL and newborn care; (4) PPFP choice.

Working diagnosis: always state *stage and phase of labour*, *parity*, *gestation*, *presentation/position*, and *maternal and fetal condition*: "Multigravida at term in the active first stage, cephalic LOA, mother and fetus in good condition."

EXAM FAVOURITE

Worked example 1: is she in labour? A 20-year-old primigravida at 40 weeks has had contractions for 6 hours. At 10:00: 2 contractions in 10 min lasting 20 s, cervix 2 cm, 30% effaced, head 4/5. At 14:00: 2 contractions in 10 min, cervix still 2 cm. **Interpretation**: no cervical change over 4 hours with weak, infrequent contractions = **false labour or early latent phase**. Mother and fetus well → do not start a partograph, do not augment; give analgesia and fluids, and either observe or discharge home with danger signs and a plan to return.

EXAM FAVOURITE

Worked example 2: reassessing progress. A P3 was admitted at 4 cm at 08:00 (head 3/5, 3 contractions/10 min). At 12:00 the cervix is 5 cm, head 3/5, **moulding ++, caput ++**, contractions 4 in 10 min lasting 50 s, bladder full. **Interpretation**: only 1 cm in 4 hours, the partograph is past the **alert line** and heading for the action line; contractions are **adequate**, so poor powers is not the cause; a **multipara** with good contractions, no descent and increasing moulding = **suspect CPD/obstruction**. Action: catheterise, rehydrate, full senior assessment; **no oxytocin**; prepare for caesarean section if descent does not improve (at an HC III: refer now).

Model presentation (active labour)

"This is **Mrs N.A., a 24-year-old G2P1+0** at **39 weeks by dates consistent with an early scan**, admitted at 06:00 with **regular painful contractions since 01:00** and a show. Her ANC was attended four times; she is **HIV-negative**, VDRL non-reactive, Hb 11.8 g/dL, blood group O positive. Her previous delivery was a spontaneous vaginal delivery of a 3.2 kg baby two years ago.

On examination she is in good general condition, afebrile, **pulse 88, BP 118/72**. The fetus is in **longitudinal lie, cephalic presentation, LOA, 2/5 palpable**, with **3 contractions in 10 minutes lasting 40 seconds**, FHR 140 bpm regular. On VE at 06:30 the **cervix was 5 cm, 80% effaced**, membranes intact, vertex at station -1, **no caput, no moulding**, adequate pelvis.

Summary: a multigravida at term in the **active first stage of labour**, mother and fetus in good condition. **Plan**: partograph/LCG started, supportive care with companion and oral fluids, FHR and contractions every 30 minutes, VE at 10:30 or earlier if indicated, prepare for AMTSL and newborn care."

6. Common Problems and Their Management

First-5-minute emergencies on admission

If you find…	Do this now (then call a senior)
Head visible / urge to push	Move to the delivery couch, open the pack, draw up **oxytocin 10 IU**, prepare the newborn corner
Bleeding more than a show	**No digital VE**; two large-bore IV lines, Hb, group and crossmatch; think abruption, praevia, rupture
BP ≥160/110, or fits	$MgSO_4$ loading dose and an antihypertensive per protocol (see the pre-eclampsia topic)
FHR <110 or >160	Left lateral, IV fluids, stop any oxytocin, VE to exclude cord prolapse, call a senior
Cord at the introitus / felt on VE	Knee-chest or exaggerated Sims position, push the presenting part up, fill the bladder, emergency CS
Temperature ≥38°C, pulse >120	Look for chorioamnionitis/malaria/sepsis; cultures if possible; antibiotics
Shock index ≥0.9, pale, collapsed	ABC, two IV lines, fluids, think rupture or abruption

Emergency flow: something goes wrong during a "normal" labour

1. **Call for help** (senior midwife/doctor).
2. **ABC for the mother**: left lateral position, oxygen if hypoxic, IV access with two large-bore cannulas, send blood for Hb/grouping/crossmatch.
3. **Assess the baby**: FHR; if abnormal, stop oxytocin, IV fluids, left lateral, VE to exclude cord prolapse and assess progress.
4. **Decide the route and timing of delivery**: fully dilated with head low (0/5, station +2 or more) = assisted vaginal delivery (vacuum); not fully dilated or head high = emergency caesarean.
5. **If at HC III**: stabilise, refer with a completed referral note and the partograph, accompanying health worker, IV line running, and (if relevant) $MgSO_4$ loading dose or first-dose antibiotics already given.

Slow progress

- Recheck the 3 Ps in order: count contractions, empty the bladder, rehydrate; check position (OP, deflexion), fifths vs station, moulding and caput.
- **Primigravida with inadequate contractions and no signs of obstruction**: senior review for augmentation where CS is available.
- **Multipara with good contractions**: obstruction until proved otherwise; **no oxytocin**; catheterise, rehydrate, antibiotics, prepare for CS.
- **At an HC III**: refer when the alert line is crossed.

Shoulder dystocia

> **RED FLAG**
> **Shoulder dystocia**: the head is born and then retracts against the perineum ("turtle sign"), and the shoulders do not deliver with gentle traction. Call for help, stop pushing, do not pull hard, **McRoberts manoeuvre + suprapubic pressure** (HELPERR). See the vaginal delivery skills topic.

Retained placenta

> RED FLAG
> **Retained placenta**: placenta not delivered 30 minutes after birth despite AMTSL. Empty the bladder, repeat oxytocin 10 IU IM, try CCT again; if still retained, **manual removal of placenta** under analgesia/anaesthesia with IV antibiotics (single dose ampicillin 2 g IV + metronidazole 500 mg IV, or cefazolin 1 g IV + metronidazole 500 mg IV). If bleeding, manage as PPH.

Postpartum haemorrhage

- Blood loss **≥500 mL is PPH** and triggers the **E-MOTIVE** bundle (early detection with a calibrated drape; uterine massage, oxytocic drugs, tranexamic acid, IV fluids, examination and escalation); see the PPH topic.
- Think of the **4 Ts**: tone, trauma, tissue, thrombin.

Complications of labour and how to prevent them

Complication	Cause/mechanism	Prevention
Prolonged/obstructed labour	CPD, malposition, malpresentation	Partograph/LCG, timely referral, CS
Uterine rupture	Obstruction, oxytocin misuse in multiparas, previous CS	Never augment a multipara without excluding obstruction; TOLAC only in hospital
Postpartum haemorrhage	Atony (tone), trauma, retained tissue, coagulopathy (the 4 Ts)	AMTSL, monitoring in the fourth stage, early detection
Perineal tears/OASI	Uncontrolled delivery of the head, instrumental delivery, big baby	Controlled crowning, warm compress, restrictive episiotomy
Infection (chorioamnionitis, puerperal sepsis)	Prolonged ROM, many VEs, poor asepsis	Hand hygiene, limit VEs, antibiotics when indicated
Fetal hypoxia/birth asphyxia	Tachysystole, cord compression, prolonged labour	FHR monitoring, avoid over-augmentation, prompt delivery
Obstetric fistula (VVF/RVF)	Prolonged pressure necrosis of the bladder/rectum by the impacted head	Prevent obstructed labour; catheterise for 10-14 days after prolonged obstructed labour
Uterine inversion	Cord traction without counter-traction on a relaxed uterus	Always counter-traction during CCT
Shoulder dystocia	Macrosomia, diabetes, prolonged second stage	Anticipate, HELPERR drill

7. Guidelines, Policy and Law in Uganda

- **Uganda Clinical Guidelines 2023**: modified WHO partograph (active phase from 4 cm), AMTSL with oxytocin 10 IU IM, **misoprostol 600 micrograms orally** where oxytocin is unavailable.
- **Essential Maternal and Newborn Clinical Care Guidelines (2022)**: adopt the **WHO Labour Care Guide** (active phase from 5 cm).
- **WHO 2018 intrapartum care for a positive childbirth experience**: respectful maternity care, birth companion, no routine interventions (enema, shaving, amniotomy, IV fluids, oxytocin, CTG), upright positions, restrictive episiotomy, second stage up to about 3 h (nullipara) and 2 h (multipara).
- **Chlorhexidine 7.1% gel cord care**: Uganda policy since **2016**.
- **Viability threshold**: Uganda uses **28 weeks** for the definition of labour; WHO uses 22 weeks for statistics.

- **Facility discharge**: at least **24 hours** after a facility birth (WHO).

UGANDA CONTEXT

Where Uganda and WHO/international guidance diverge:

- **Active phase**: modified WHO partograph and UCG **4 cm**; WHO 2018 and the LCG **5 cm**. Say which you are using.
- **Misoprostol for PPH prevention**: UCG **600 micrograms orally**; WHO allows **400 or 600 micrograms**.
- **Heat-stable carbetocin** is a WHO-recommended alternative but is supplied in only some Ugandan units.
- **Temperature monitoring**: modified partograph every 2 h; LCG every 4 h.
- **Antenatal steroids**: these notes use 28–34 weeks (Ugandan practice); WHO 2022 recommends them from 24 to 34 weeks where gestational age is reliable and preterm care is adequate Verify *current UCG gestation window for antenatal corticosteroids*.

UGANDA CONTEXT

Women living with HIV: continue their ART in labour (TLD), avoid artificial rupture of membranes and unnecessary episiotomy or instrumental delivery where possible (they increase the baby's exposure to maternal blood/secretions), and start infant prophylaxis (nevirapine +/- zidovudine per the current consolidated guidelines) within 6-12 hours of birth.

8. Key Points

High-yield points

- **Labour = regular painful contractions + cervical change.**
- Active phase: **4 cm** (modified WHO partograph) / **5 cm** (WHO 2018, LCG).
- First 5 minutes: pulse, BP, temperature, FHR after a contraction; bleeding, liquor, movements, urge to push.
- **Fifths next to station**; 3/5 with "+2" is caput.
- **Slow progress in a multipara = obstruction**; no oxytocin.
- **Suboccipitobregmatic 9.5 cm**: flexion makes OA labour normal.
- **Every birth gets oxytocin 10 IU IM within 1 minute**, after excluding a second twin.
- Misoprostol **600 micrograms** orally if no oxytocin; **no ergometrine** in hypertension.
- Newborn: dry, skin-to-skin, delayed cord clamping, breastfeed within 1 hour, vitamin K, eye ointment, chlorhexidine.
- Fourth stage: tone, bleeding, pulse, BP every 15 minutes for 2 hours.

Numbers & doses

Item	Value
Term	37+0 to 41+6 weeks
Active phase starts	**4 cm** (modified WHO partograph) / **5 cm** (WHO 2018, LCG)
Alert line / action line	Starts at 4 cm, **1 cm/h**; action line **4 h to the right**
LCG cervix alert thresholds	**6 h at 5 cm, 5 h at 6 cm, 3 h at 7 cm, 2.5 h at 8 cm, 2 h at 9 cm**
Adequate contractions	**3–5 in 10 min**, each **40–60 s**; >5 in 10 min = tachysystole
Normal FHR	**110–160 bpm**; every 30 min in first stage, every 5 min in second stage

Item	Value
VE frequency	**Every 4 h** in the active first stage (sooner only if indicated)
Maternal observations (partograph)	Pulse 30-min, BP 4-h, temperature 2-h, urine every void
Active first stage duration	Median about **4 h** (nullipara, ≤12 h) / **3 h** (multipara, ≤10 h)
Second stage (WHO 2018)	Up to about **3 h** nullipara, **2 h** multipara
Third stage	Prolonged/retained placenta if **>30 min**
Prolonged rupture of membranes	**>18 h**
Shock index	Pulse ÷ SBP **≥0.9** abnormal
Pethidine	**50–100 mg IM** with an antiemetic; avoid within 2–3 h of expected birth
AMTSL uterotonic	**Oxytocin 10 IU IM within 1 min** (after excluding a second twin)
No oxytocin available	**Misoprostol 600 mcg orally**
Carbetocin / ergometrine	**100 mcg** IM/IV / **0.2 mg** IM
Manual removal antibiotics	Ampicillin **2 g IV** + metronidazole **500 mg IV** (or cefazolin 1 g IV + metronidazole)
Fourth-stage checks	Tone, bleeding, pulse, BP **every 15 min for 2 h**
Normal blood loss	**<500 mL**
Delayed cord clamping	**1–3 min** (not before 1 min) in a vigorous baby
Vitamin K	**1 mg IM** (0.5 mg preterm/LBW)
BCG / OPV-0	**0.05 mL** intradermal / **2 drops** oral at birth
Antenatal steroids	Dexamethasone **6 mg IM 12-hourly × 4** (28–34 weeks)
Well-flexed vertex diameter	Suboccipitobregmatic **9.5 cm**
Engagement	**≤2/5** of the head palpable abdominally
Discharge after facility birth	At least **24 h**

Exam traps

- **"She has a show and ruptured membranes, so she is in labour."** → Wrong. Labour = **regular painful contractions + cervical change**. A show and ROM support it but do not define it.
- **"Slow progress in a multipara: start oxytocin."** → Wrong. Poor powers are uncommon in a multipara; slow progress is **obstruction until proved otherwise**, and oxytocin can rupture her uterus.
- **"Station +2 on VE, so the head is low."** → Only if the abdomen agrees. **3/5 palpable with "station +2" = caput**, not descent.
- **"Plot the first VE at 2 cm on the partograph."** → Wrong. The partograph starts in the **active phase** (4 cm modified WHO / 5 cm LCG); latent-phase findings go in the notes.
- **"Give oxytocin as soon as the baby is born."** → Almost: first **palpate the abdomen to exclude a second twin**, then give it within 1 minute.
- **"Ergometrine is a good alternative to oxytocin for everyone."** → Wrong. **Contraindicated in hypertension, pre-eclampsia and cardiac disease**; misoprostol 600 mcg orally is the Uganda alternative.
- **"Suction every baby's mouth and nose at birth."** → Wrong. **Dry first**; suction only if secretions block the airway.
- **"Cervix dilating at less than 1 cm/h means she needs augmentation."** → WHO 2018: slower dilatation **alone** is not an indication to intervene if mother and baby are well; look for a cause first.

Questions seniors ask

Viva questions

1. **What are the signs of placental separation?** Uterus becomes firm, globular and rises; gush of blood; lengthening of the cord at the vulva.
2. **Why is the suboccipitobregmatic diameter important?** It is the smallest AP diameter (9.5 cm), presented by a well-flexed vertex, so normal labour requires flexion.
3. **What is AMTSL and why does it work?** Uterotonic within 1 min, CCT with counter-traction, uterine tone checks. The oxytocin makes the uterus contract, which shears the placenta off and clamps the spiral arteries ("living ligatures").
4. **When would you not give ergometrine?** Hypertension/pre-eclampsia, cardiac disease, retained second twin; it causes vasoconstriction and a sustained tetanic contraction.
5. **Define engagement and how you assess it.** Passage of the biparietal diameter through the inlet; 2/5 or less palpable abdominally.
6. **What is the Ferguson reflex?** Stretch of the cervix/upper vagina by the presenting part leads to oxytocin release, reinforcing contractions.
7. **Name the components of the partograph.** Fetal condition (FHR, liquor, moulding), labour progress (cervix, descent, contractions), drugs and fluids, maternal condition (pulse, BP, temperature, urine).
8. **What is restitution?** The return of the head to its natural alignment with the shoulders after delivery of the head (a 45-degree turn back towards the side where the occiput started).

On the ward round

1. **"Is she in labour, and how do you know?"** Regular painful contractions (e.g. 3 in 10 min lasting 40 s) with cervical change on two examinations, or a cervix already ≥4–5 cm and effaced.
2. **"Why do you check the abdomen before the VE?"** To know lie, presentation and **descent in fifths**, so you can interpret station correctly and not mistake caput for descent.
3. **"What will make you call me during this labour?"** FHR <110 or >160 or late decelerations, thick meconium, bleeding, fever ≥38°C, BP ≥160/110 or fits, crossing the action line (or an LCG alert), signs of obstruction, or a cord prolapse.
4. **"What is the first thing you do after the baby is born?"** Dry and assess breathing; palpate for a second twin; **oxytocin 10 IU IM within 1 minute**.
5. **"The placenta is not out after 30 minutes. What now?"** Empty the bladder, repeat oxytocin 10 IU IM, retry CCT; if still retained, manual removal under analgesia with prophylactic antibiotics; treat any bleeding as PPH.
6. **"Why does a full bladder matter in labour?"** It impedes descent in the first and second stages and causes uterine atony (PPH) after birth.
7. **"How much blood did she lose, and how do you know?"** Estimate by a calibrated drape or by weighing pads; **≥500 mL is PPH** and triggers the E-MOTIVE bundle.

Mnemonics

- **Cardinal movements**: "**E**very **D**ecent **F**etus **I**s **E**xceptionally **R**otated **E**very **E**xpulsion" = Engagement, Descent, Flexion, Internal rotation, Extension, Restitution, External rotation, Expulsion.
- **Signs of placental separation**: "**G**ush, **L**engthening, **R**ise" (GLR) of the uterus.
- **AMTSL**: "**O**-**C**-**M**": Oxytocin in 1 minute, Controlled cord traction, Massage/tone checks.
- **Progress problems**: the "3 Ps" (Powers, Passenger, Passage) plus Psyche.
- **Newborn care**: "**D**ry, **S**kin-to-skin, **D**elay cord clamp, **B**reastfeed, **V**itamin K, **E**yes, **C**ord care" = "Do Some Deliveries, Babies Very Easily Cry".
- **Pelvis**: "**In**let **T**ransverse, **Out**let **A**P": the head enters sideways and leaves face-down.

OSCE tips

- Mechanism of labour on a mannequin: narrate each movement *and* say what the pelvis is doing (for example, "the head flexes as it meets resistance, presenting the 9.5 cm diameter").
- AMTSL station: say "I palpate the abdomen to exclude a second twin" *before* you give oxytocin. That single sentence often carries marks.
- Always greet, consent, maintain dignity and cover the woman.
- Obstetric abdominal examination: inspection, fundal height, lie, presentation, position, engagement in fifths, contractions, then the fetal heart after a contraction; then say you would do a VE with a chaperone.

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SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define labour.	Regular, painful uterine contractions that cause **progressive cervical effacement and dilatation**, ending in expulsion of the fetus, placenta and membranes.
2	When does the active first stage begin (WHO 2018 vs modified WHO partograph)?	**WHO 2018 / Labour Care Guide: 5 cm.** Modified WHO partograph (still used in many Ugandan units): **4 cm**.
3	List the 8 cardinal movements of labour in order.	Engagement, Descent, Flexion, Internal rotation, Extension, Restitution, External rotation, Expulsion.
4	Which diameter does a well-flexed vertex present and how long is it?	**Suboccipitobregmatic, 9.5 cm** (the smallest AP diameter).
5	Give the diameters presented by a deflexed (OP) head, a brow and a face.	Deflexed vertex: **occipitofrontal 11.5 cm**. Brow: **mentovertical 13 cm** (undeliverable at term). Face: **submentobregmatic 9.5 cm**.
6	Clinical definition of engagement?	The biparietal diameter has passed the pelvic inlet; abdominally **2/5 or less of the head palpable** above the brim.
7	Why does the fetal head rotate internally?	The **gutter-shaped levator ani** directs the leading occiput forwards, and the **outlet is widest anteroposteriorly**, so the head turns 45 degrees from LOA to OA.
8	What is the Ferguson reflex?	Stretching of the cervix/upper vagina by the presenting part triggers **posterior pituitary oxytocin release**, which strengthens contractions (positive feedback).
9	Three components of AMTSL?	1. **Uterotonic within 1 minute** of birth (oxytocin 10 IU IM). 2. **Controlled cord traction** with counter-traction. 3. **Uterine tone assessment/massage** after the placenta, every 15 min for 2 h.
10	Uterotonic for AMTSL if oxytocin is unavailable (Uganda)?	**Misoprostol 600 micrograms orally**, once. (WHO accepts 400-600 mcg; heat-stable carbetocin 100 mcg IM/IV where supplied.)
11	Contraindications to ergometrine?	**Hypertension / pre-eclampsia**, cardiac disease, undelivered second twin (and caution in retained placenta). It causes vasoconstriction and a sustained tonic contraction.
12	Three signs of placental separation?	1. Uterus becomes **firm, globular and rises**. 2. **Gush of blood**. 3. **Lengthening of the cord** at the vulva.
13	What is the purpose of counter-traction during CCT?	Pushing the uterus up above the symphysis while pulling on the cord **prevents uterine inversion**.
14	Normal duration of the second stage (WHO 2018)?	Usually **up to 3 h in first labours** and **up to 2 h in subsequent labours**, provided mother and baby are well.
15	How often do you auscultate the FHR in the active first stage and the second stage?	Active first stage: **every 30 min** (WHO: every 15-30 min), for a full minute after a contraction. Second stage: **every 5 min** (or after each contraction).
16	How often should a VE be done in a normal active first stage?	**Every 4 hours**, or sooner if clinically indicated (e.g. urge to push, abnormal FHR, rupture of membranes).
17	When should the cord be clamped in a vigorous newborn?	**Delayed: 1-3 minutes** after birth (not earlier than 1 minute). Clamp immediately only if the baby needs ventilation and cannot be ventilated with the cord intact.

#	Question	Answer
18	Routine newborn drugs after birth in Uganda?	**Vitamin K 1 mg IM** (0.5 mg preterm), **tetracycline 1% eye ointment** once, **chlorhexidine 7.1% gel** to the cord stump; plus BCG and OPV-0.
19	Define a retained placenta and first steps.	Placenta not delivered **30 minutes** after birth. Empty bladder, repeat **oxytocin 10 IU IM**, repeat CCT; if still retained, **manual removal** with analgesia and IV antibiotics.
20	What does descent in fifths tell you that station does not?	It is **not fooled by caput**. A head 3/5 palpable abdominally is not engaged, even if the VE feels something at +2 (that is caput).
21	Grades of moulding?	**0** bones separate; **+** touching; **++** overlapping but reducible; **+++** overlapping, not reducible (suggests CPD).
22	Four monitoring questions to ask every time you review a labouring woman?	Is the **mother** well? Is the **baby** well? Is labour **progressing**? Is the **partograph/LCG** up to date (and is there an alert)?
23	What do you check in the first 5 minutes when a woman arrives in labour?	**Look** (pushing, bleeding, fitting, cord at introitus), **vitals** (pulse, BP, temperature, RR, shock index), **FHR** for 1 min after a contraction, and ask about **bleeding, liquor colour, fetal movements and urge to push**.
24	Admission VE: head 3/5 palpable abdominally but station +2 with caput ++. Interpretation?	The "low" head is **caput**, not descent. The head is not engaged; with moulding this suggests **CPD/obstruction**. Senior review, no oxytocin.
25	Contractions 2 in 10 min for 20 s; cervix 2 cm unchanged after 4 h; mother and fetus well. Diagnosis and plan?	**False labour / early latent phase.** No partograph, no augmentation; analgesia, fluids, observe or discharge with danger signs.
26	Which admission tests do you check or do for a woman in labour (Uganda)?	**Hb (HemoCue)**, blood group and Rh, **HIV and syphilis RDT** if unknown/retest due, **urine dipstick** (protein, ketones), malaria RDT if febrile, glucose if indicated; ultrasound only if presentation, number or placenta site is uncertain.
27	Bedside clotting test: how is it read?	2 mL of venous blood in a plain glass tube, kept warm and tilted: **no clot by 7 minutes, or a soft clot that breaks down easily = coagulopathy** (think abruption, IUFD, massive PPH).

SELF-TEST

Single best answer questions

16 questions. The answers and explanations follow at the end.

1. A 22-year-old primigravida at 39 weeks presents to an HC IV with lower abdominal pains for 6 hours. Contractions are 2 in 10 minutes lasting 20 seconds. The cervix is 2 cm, 30% effaced, and 3 hours later it is unchanged. FHR is 140 bpm and she is otherwise well. What is the MOST appropriate management?

A Supportive care, oral fluids, analgesia and reassessment; she may walk and go home if all is well
B Perform artificial rupture of membranes
C Start oxytocin augmentation
D Refer urgently to a regional referral hospital for caesarean section
E Give IM pethidine and start a partograph at 2 cm

2. Immediately after a normal vaginal birth of a 3.1 kg baby at an HC III, the midwife is about to give oxytocin. What must be done FIRST?

A Clamp and cut the cord
B Give misoprostol 600 micrograms orally
C Apply controlled cord traction
D Palpate the abdomen to exclude another fetus
E Massage the uterus vigorously

3. A 29-year-old P1 has just delivered. She has chronic hypertension and her BP is 158/102 mmHg. There is no oxytocin in the facility's fridge today, but misoprostol and ergometrine are available. Which uterotonic is MOST appropriate for AMTSL?

A Ergometrine 0.2 mg IM
B Misoprostol 600 micrograms orally
C Oxytocin-ergometrine combination IM
D No uterotonic; physiological third stage
E Misoprostol 1000 micrograms rectally

4. A 30-year-old P5 is admitted at 6 cm. Four hours later she is still 6 cm despite 4 strong contractions in 10 minutes lasting 50 seconds. The head is 3/5 palpable, moulding is +++ and there is a large caput. Her urine is dark. What is the MOST appropriate next step?

A Start oxytocin infusion to strengthen contractions
B Rupture membranes and review in 2 hours
C Apply a vacuum extractor
D Encourage her to push
E Prepare for caesarean section (refer if at a facility without theatre) after catheterisation, IV fluids and antibiotics

5. During the second stage, the head of the baby is delivered. The occiput then turns spontaneously towards the mother's left thigh. What is this movement called and what does it indicate?

A Internal rotation; the shoulders are entering the pelvis
B External rotation; the posterior shoulder has delivered
C Restitution; the head is realigning with the shoulders, the baby was LOA
D Extension; the occiput is pivoting under the symphysis
E Restitution; the baby was ROA

6. A primigravida has been fully dilated for 1 hour with the head at 0/5 and good contractions. The FHR is 145 bpm with no decelerations, and she is pushing well. The student proposes an episiotomy 'to speed things up'. What is the BEST response?

A Disagree; continue supportive care with perineal protection; episiotomy only for a clear indication
B Agree; a midline episiotomy reduces third-degree tears
C Agree; episiotomy is routine for primigravidae
D Disagree; start oxytocin instead
E Disagree; apply fundal pressure to shorten the second stage

7. Twenty minutes after delivery with AMTSL, the placenta has not delivered. The uterus is well contracted and there is minimal bleeding. The bladder is palpable. What is the MOST appropriate next step?

A Proceed immediately to manual removal under general anaesthesia
B Give ergometrine 0.5 mg IV
C Pull harder on the cord
D Empty the bladder and repeat controlled cord traction with the next contraction
E Refer to a regional referral hospital

8. A woman in active labour has an abdominal exam showing the head 3/5 palpable. On VE the student reports 'station +2'. There is a large caput. How should these findings be interpreted? Advanced

A The head is engaged and delivery is imminent
B The student is feeling caput; the head is not engaged
C The abdominal exam is wrong because it is less accurate
D The fetus is in breech presentation
E There is placenta praevia

9. A baby is born at term, cries immediately and has good tone. What is the recommended timing of cord clamping?

A Immediately, to reduce polycythaemia
B Within 30 seconds
C Only after the first breastfeed
D After the placenta is delivered
E 1 to 3 minutes after birth

10. A P2 at term is 7 cm dilated. Suddenly after spontaneous rupture of membranes, the FHR drops to 80 bpm and stays there for 3 minutes. What is the MOST important immediate action?

A Start oxytocin to speed up delivery
B Repeat FHR in 30 minutes
C Perform a vaginal examination to check for cord prolapse
D Give oral fluids
E Allow her to push

11. During CCT, the midwife pulls on the cord without supporting the uterus, and the fundus is no longer palpable abdominally. The woman is suddenly shocked with pulse 130 and BP 80/40, with modest bleeding. What is the MOST likely diagnosis? Advanced

A Acute uterine inversion
B Amniotic fluid embolism
C Uterine atony
D Ruptured uterus
E Cervical tear

12. A primigravida at term is 5 cm dilated. Membranes ruptured 2 hours ago with clear liquor. She is anxious, hungry and wants to walk around. What is the BEST advice?

A Nil by mouth and bed rest in supine position
B Give an enema and shave the perineum
C Start routine IV fluids and oxytocin
D Allow light food and oral fluids, encourage mobility and upright positions, support with a companion
E Continuous CTG monitoring

13. Examining a placenta after delivery, you see a vessel running across the membranes to a torn edge with no placental tissue attached. The uterus is contracted and bleeding is currently normal. What is the MOST appropriate action? Advanced

A No action; this is a normal finding
B Suspect a retained succenturiate lobe; explore the uterus (manual/instrumental evacuation) with antibiotics and uterotonic cover
C Discharge with oral antibiotics
D Give misoprostol and observe for 6 hours only
E Perform an ultrasound in 6 weeks

14. At an HC III, a primigravida's partograph crosses the alert line at 6 cm. FHR is normal and she has 3 moderate contractions in 10 minutes. The nearest HC IV is 2 hours away. What is the BEST action?

A Wait until she reaches the action line, then refer
B Start oxytocin at the HC III
C Tell the family to find their own transport tomorrow
D Perform ARM and wait 4 hours
E Refer now to the HC IV, with the partograph, IV line and escort

15. A 26-year-old woman living with HIV on TLD, with a suppressed viral load, is in active labour at 6 cm with intact membranes and normal progress. Which intrapartum practice is MOST appropriate?

A Routine artificial rupture of membranes to speed labour
B Elective episiotomy to shorten the second stage
C Continue ART in labour, avoid unnecessary ARM, episiotomy and instrumental delivery, and give infant prophylaxis after birth
D Stop ART in labour to avoid drug toxicity in the baby
E Immediate caesarean section for all women with HIV

16. A student is asked why a brow presentation at term usually cannot deliver vaginally while a well-flexed vertex can. What is the BEST explanation?

A The brow presents the mentovertical diameter (13 cm), which is larger than any pelvic diameter
B The brow presents the suboccipitobregmatic diameter (9.5 cm)
C The brow cannot rotate because the pelvic floor is too strong
D The brow presents the biparietal diameter (9.5 cm)
E Brow presentation causes placental abruption

Answers & explanations

1. A · Supportive care, oral fluids, analgesia and reassessment; she may walk and go home if all is well

Irregular, short contractions with no cervical change over 3 h is **false labour or early latent phase**. WHO 2018 advises against augmentation or ARM before 5 cm if mother and baby are well. Oxytocin (C) and ARM (B) are unnecessary interventions that raise the risk of hyperstimulation, infection and caesarean. There is no indication for CS (D). The partograph/LCG is not started in the latent phase (E), and plotting at 2 cm would produce a false 'slow labour'.

2. D · Palpate the abdomen to exclude another fetus

Before any uterotonic you must **exclude an undiagnosed second twin**, because oxytocin can make the uterus contract tonically around the second baby. Cord clamping (A) should be delayed 1-3 min in a vigorous baby. CCT (C) comes after the uterotonic and a contraction. Misoprostol (B) is only for when oxytocin is unavailable. Vigorous massage before the placenta is out (E) can cause partial separation and bleeding.

3. B · Misoprostol 600 micrograms orally

With no oxytocin, the Uganda Clinical Guidelines recommend **misoprostol 600 mcg orally**. Ergometrine and the oxytocin-ergometrine combination (A, C) are **contraindicated in hypertension** because they cause vasoconstriction and can precipitate stroke or eclampsia. Omitting a uterotonic (D) doubles PPH risk. 1000 mcg rectal (E) is an old PPH *treatment* regimen, not a prevention dose.

4. E · Prepare for caesarean section (refer if at a facility without theatre) after catheterisation, IV fluids and antibiotics

Strong contractions without progress, a high head, severe moulding and caput in a grand multipara = **obstructed labour**. Management is resuscitation (IV fluids, catheter, antibiotics) and **caesarean section**. Oxytocin (A) in an obstructed multipara can **rupture the uterus**. ARM (B) delays delivery. Pushing (D) before full dilatation is useless and harmful. Vacuum (C) needs full dilatation and a head at 0/5 to 1/5.

5. C · Restitution; the head is realigning with the shoulders, the baby was LOA

Restitution is the 45-degree untwisting of the neck after the head is born; the occiput returns to the side it came from, so a turn to the mother's left means **LOA**. Internal rotation (A) happens inside the pelvis before crowning. External rotation (B) follows restitution as the anterior shoulder rotates forwards; the shoulders have not delivered yet. Extension (D) occurs as the head is born. ROA (E) would restitute to the right.

6. A · Disagree; continue supportive care with perineal protection; episiotomy only for a clear indication

WHO 2018 recommends against **routine** or liberal episiotomy. Mother and baby are well and the second stage is within normal limits, so continue with warm compresses and a controlled delivery of the head. Routine episiotomy (C) increases blood loss and pain without benefit. **Midline episiotomy** (B) actually increases the risk of OASI. Oxytocin (D) is not indicated with good contractions. **Fundal pressure** (E) is not recommended; it can cause uterine rupture and fetal injury.

7. D · Empty the bladder and repeat controlled cord traction with the next contraction

At 20 min the placenta is not yet 'retained' (the cut-off is **30 min**). A full bladder can prevent descent, so **catheterise/empty the bladder** and retry CCT with counter-traction. Manual removal (A) is for a placenta still retained after 30 min or with bleeding. Pulling harder (C) risks cord avulsion and **inversion**. Ergometrine IV (B) can clamp the cervix and trap the placenta; the dose is also wrong. Referral (E) is premature: an HC III can do manual removal.

8. B · The student is feeling caput; the head is not engaged

Abdominal descent in fifths is **not fooled by caput**. 3/5 palpable means the head is not engaged, so the '+2' is swelling of the scalp. This discrepancy, especially with moulding, suggests **CPD**. Option A ignores the abdominal finding; C is backwards (descent in fifths is the more reliable measure when there is caput). Breech (D) would give a soft, irregular presenting part and a head at the fundus. Praevia (E) is not suggested and a VE should not be done if it were suspected.

9. E · 1 to 3 minutes after birth

WHO recommends **delayed cord clamping, not earlier than 1 minute (1-3 min)** for vigorous babies. It transfers about 80-100 mL of blood, improving iron stores and reducing anaemia in infancy, which matters a lot in Uganda. Immediate clamping (A, B) loses this benefit; polycythaemia is rarely significant. Waiting for the placenta (D) or after breastfeeding (C) is not the standard and can delay care.

10. C · Perform a vaginal examination to check for cord prolapse

Sudden prolonged bradycardia after membrane rupture = **cord prolapse until proven otherwise**, so do an **immediate VE**. If present: call for help, elevate the presenting part, knee-chest or Trendelenburg position, fill the bladder, and arrange emergency caesarean. Oxytocin (A) would worsen hypoxia. Waiting 30 min (B) could lose the baby. Oral fluids (D) do nothing for acute hypoxia. Pushing (E) is inappropriate at 7 cm.

11. A · Acute uterine inversion

Traction on the cord without **counter-traction**, a fundus that cannot be felt, and shock **out of proportion to blood loss** (vagal/neurogenic shock) are classic for **acute uterine inversion**. Atony (C) gives a soft, large, palpable uterus. AFE (B) causes collapse, hypoxia and DIC, not a missing fundus. Rupture (D) is usually in labour, often with previous scars or obstruction. A cervical tear (E) causes bleeding with a contracted uterus.

12. D · Allow light food and oral fluids, encourage mobility and upright positions, support with a companion

WHO 2018 supports **oral fluids and food** in low-risk labour, **mobility and upright positions**, and a **birth companion**. Supine rest (A) causes aortocaval compression and is less comfortable; routine starvation causes ketosis. Routine IV fluids and oxytocin (C) are not recommended. Enema and shaving (B) are outdated with no benefit. Continuous CTG (E) is not recommended for low-risk women and increases caesarean rates without improving outcomes.

13. B · Suspect a retained succenturiate lobe; explore the uterus (manual/instrumental evacuation) with antibiotics and uterotonic cover

Vessels running to a torn membrane edge suggest a **succenturiate (accessory) lobe left behind**, which will cause PPH or sepsis. The uterus must be explored (manual exploration/evacuation) under analgesia with antibiotics and oxytocin. It is not normal (A). Oral antibiotics alone (C), watchful waiting (D) or a delayed scan (E) leave retained tissue that can bleed catastrophically.

14. E · Refer now to the HC IV, with the partograph, IV line and escort

At a facility without CS and blood, crossing the **alert line is the trigger to refer** because by the time she reaches the action line she may already be obstructed and transport takes time. Waiting (A) and ARM-and-wait (D) delay care. Oxytocin augmentation (B) needs close monitoring and the ability to perform CS if it fails, so it should be done at the HC IV/hospital. Option C ignores the second delay of the three delays model.

15. C · Continue ART in labour, avoid unnecessary ARM, episiotomy and instrumental delivery, and give infant prophylaxis after birth

For eMTCT, **continue ART**, minimise procedures that expose the baby to maternal blood and secretions (ARM, episiotomy, instrumental delivery, fetal scalp procedures) and start **infant prophylaxis** soon after birth. Routine ARM (A) and routine episiotomy (B) increase exposure. Stopping ART (D) risks viral rebound and transmission. Caesarean (E) is not indicated for HIV alone with a suppressed viral load.

16. A · The brow presents the mentovertical diameter (13 cm), which is larger than any pelvic diameter

A brow presents the **mentovertical diameter (about 13 cm)**, which exceeds the pelvic diameters, so labour obstructs unless the head flexes or extends fully. The suboccipitobregmatic (B) is presented by a well-flexed vertex. Pelvic floor strength (C) is not the reason. The biparietal diameter (D) is transverse and is the same in all cephalic presentations. Abruption (E) is unrelated.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: A primigravida in early labour at Mengo

Ms A.K., 21 years, G1P0 at 39+2 weeks by an early scan, walks into Mengo labour ward at 22:00 with painful tightenings since 18:00 and a blood-stained mucus show. She attended ANC 5 times; HIV negative, Hb 12.1 g/dL, blood group B positive.

1. What focused history do you take and why?

> Onset, frequency and strength of contractions (true vs false labour); show, **ruptured membranes and liquor colour** (infection risk, meconium); **vaginal bleeding** (APH, avoid VE); **fetal movements**; symptoms of pre-eclampsia and infection; ANC results (HIV, syphilis, Hb, group); medical history; birth plan and companion.

2. Examination: T 36.8, P 84, BP 116/70. Cephalic, LOA, 4/5 palpable, 2 contractions in 10 min lasting 20-25 s, FHR 138 after a contraction. VE: cervix 3 cm, 50% effaced, membranes intact, station -2, no caput or moulding. What is your assessment?

> **Latent first stage** of labour (below 5 cm, WHO 2018; below 4 cm on the modified WHO partograph). Mother and fetus are well. The head is not yet engaged, which is common in labour but must be watched.

3. What is your management plan now?

> Do **not** start the partograph/LCG yet (record findings in the notes). Supportive care: companion, **oral fluids and light food**, mobility, empty bladder, simple analgesia/massage. **FHR every 30-60 min** in latent phase, reassess with VE in about 4 h or earlier if contractions become strong, membranes rupture or there is bleeding. **No oxytocin or ARM** before 5 cm if all is well.

4. At 02:30 she is 6 cm, 90% effaced, membranes ruptured with clear liquor, head 3/5, 3 contractions in 10 min lasting 40 s. How do you document and what are the monitoring frequencies?

> Start the partograph (plot 6 cm on the alert line; X for cervix, O for descent 3/5) or the LCG from 5 cm. **FHR and contractions every 30 min, pulse every 30 min, BP every 4 h, temperature every 2 h, urine each void, VE every 4 h.** Record liquor as C (clear).

5. At 06:00 she is fully dilated and pushing. How do you conduct the delivery and third stage?

> Prepare the pack, newborn corner and **oxytocin 10 IU**. Upright/semi-sitting, spontaneous pushing, **FHR every 5 min**. Warm compress; control the head at crowning (pant). Check for cord around neck, await restitution, deliver anterior then posterior shoulder. Dry baby, skin-to-skin, delay cord clamping 1-3 min. **Palpate to exclude a second twin, give oxytocin 10 IU IM within 1 min**, CCT with counter-traction, examine placenta, check tone every 15 min for 2 h, inspect perineum.

6. What newborn care and counselling do you give before discharge?

> Breastfeed within 1 h; vitamin K 1 mg IM, tetracycline eye ointment, chlorhexidine cord gel, BCG and OPV-0; no bath for 24 h; keep warm. Counsel mother on danger signs (bleeding, fever, headache, fits; baby not feeding, fast breathing, yellow, cold), exclusive breastfeeding, **postpartum family planning**, and postnatal visits (day 3, 1-2 weeks, 6 weeks). Discharge after at least 24 h.

Take-away: Diagnose the stage correctly before charting: latent phase gets supportive care, not intervention. Start the partograph/LCG in active labour, and never forget oxytocin within 1 minute after excluding a second twin.

Case 2: Slow progress in a multipara at an HC IV

Mrs S.N., 34 years, P6+0, arrives at an HC IV at 10:00 having laboured at home since the previous night. Her last baby weighed 4.1 kg. She looks tired and thirsty.

1. What key history points make you worried?

> **Grand multiparity** (P6: risk of rupture and PPH), **prolonged labour at home** (dehydration, infection, obstruction), **increasing birth weights** (macrosomia, CPD, shoulder dystocia). Ask about ROM duration, liquor, fetal movements, bleeding, fever, urine output, previous CS.

2. Exam: T 37.9, P 112, BP 110/70, dry tongue. Abdomen: cephalic, head 3/5, strong contractions 5 in 10 min lasting 55 s, a transverse groove is visible across the lower abdomen. FHR 170. VE: cervix 7 cm and oedematous, moulding +++, large caput, meconium-stained liquor, catheter urine is blood-stained. Interpretation?

> **Obstructed labour** with a **pathological retraction (Bandl's) ring** suggesting **impending uterine rupture**, maternal dehydration and possible chorioamnionitis, and **fetal distress** (FHR 170, meconium).

3. What differentials must you consider?

> CPD (big baby), malposition (OP/deflexed head), malpresentation (brow, face), hydrocephalus; complications: impending/actual uterine rupture, chorioamnionitis, abruption.

4. What is your immediate management at the HC IV?

> **Call for help.** Left lateral. **Two large-bore IV lines**, Ringer's lactate or normal saline 1 L rapidly then titrate to urine output. Blood for Hb and crossmatch (2 units). **Catheterise** (strict input-output; blood-stained urine suggests bladder injury risk). **Do NOT give oxytocin.** Broad-spectrum IV antibiotics (e.g. ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV once daily + metronidazole 500 mg IV 8-hourly). Consent and prepare for **emergency caesarean section** now.

5. What postoperative complications will you watch for?

> PPH (atony in a grand multipara), sepsis/endometritis and wound infection, **bladder injury and obstetric fistula** (keep catheter 10-14 days after prolonged obstructed labour), thromboembolism, and neonatal HIE/sepsis. Counsel on family planning (she is P7 and may want a long-acting method or tubal ligation with informed consent).

Take-away: Slow progress with strong contractions in a multipara is obstruction, not weak labour. Oxytocin can rupture her uterus. Resuscitate and deliver by caesarean.

Case 3: Third stage at an HC III without oxytocin

At an HC III at night, Ms R.T., 27 years, P1, has a spontaneous vaginal delivery of a 3.4 kg baby. The midwife finds the fridge broken and the oxytocin ampoules have been at room temperature for a week. Misoprostol tablets are available. The mother's BP was 128/80 in labour.

1. What must you do before giving any uterotonic?

> **Palpate the abdomen to exclude a second fetus.** Note the time of birth. Ensure the baby is dried, breathing, skin-to-skin.

2. Which uterotonic will you give and why?

> **Misoprostol 600 micrograms orally**, once. Oxytocin that has been out of the cold chain may have lost potency (poor quality oxytocin is a real problem in the region). Misoprostol is heat-stable. Warn about shivering and a mild fever, which are common side effects.

3. Describe how you deliver the placenta.

> Wait for a contraction and signs of separation. Clamp cord close to the perineum, **counter-traction** above the pubis pushing the uterus upwards, steady downward traction on the cord; stop if it does not descend in 30-40 s and retry with the next contraction. Deliver with both hands, twist membranes, then **examine placenta and membranes** for completeness and 3 cord vessels.

4. Fifteen minutes after the placenta, the uterus feels soft and she has soaked two pads. What do you do?

> **Call for help**, **rub up the uterus** (fundal massage) and expel clots, empty the bladder, IV access with 2 large-bore cannulas, start fluids, and follow the PPH protocol: oxytocin 20 IU in 1 L saline IV infusion if good-quality oxytocin is obtainable, otherwise misoprostol 800 micrograms sublingual; **tranexamic acid 1 g IV** over 10 min within 3 h of birth; bimanual compression; refer if not controlled. Check for tears (trauma) and retained tissue.

5. What would you document and report?

> Time of birth, uterotonic, drug and batch, placental completeness, estimated blood loss, vital signs, interventions. Report the **cold-chain failure** to the facility in-charge (drug quality is a facility-level safety issue) and discuss at the MPDSR/quality improvement meeting.

> **Take-away:** Every woman gets a uterotonic: oxytocin 10 IU IM if good quality, otherwise misoprostol 600 mcg orally. Keep checking tone for 2 hours; early detection of atony saves lives.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.